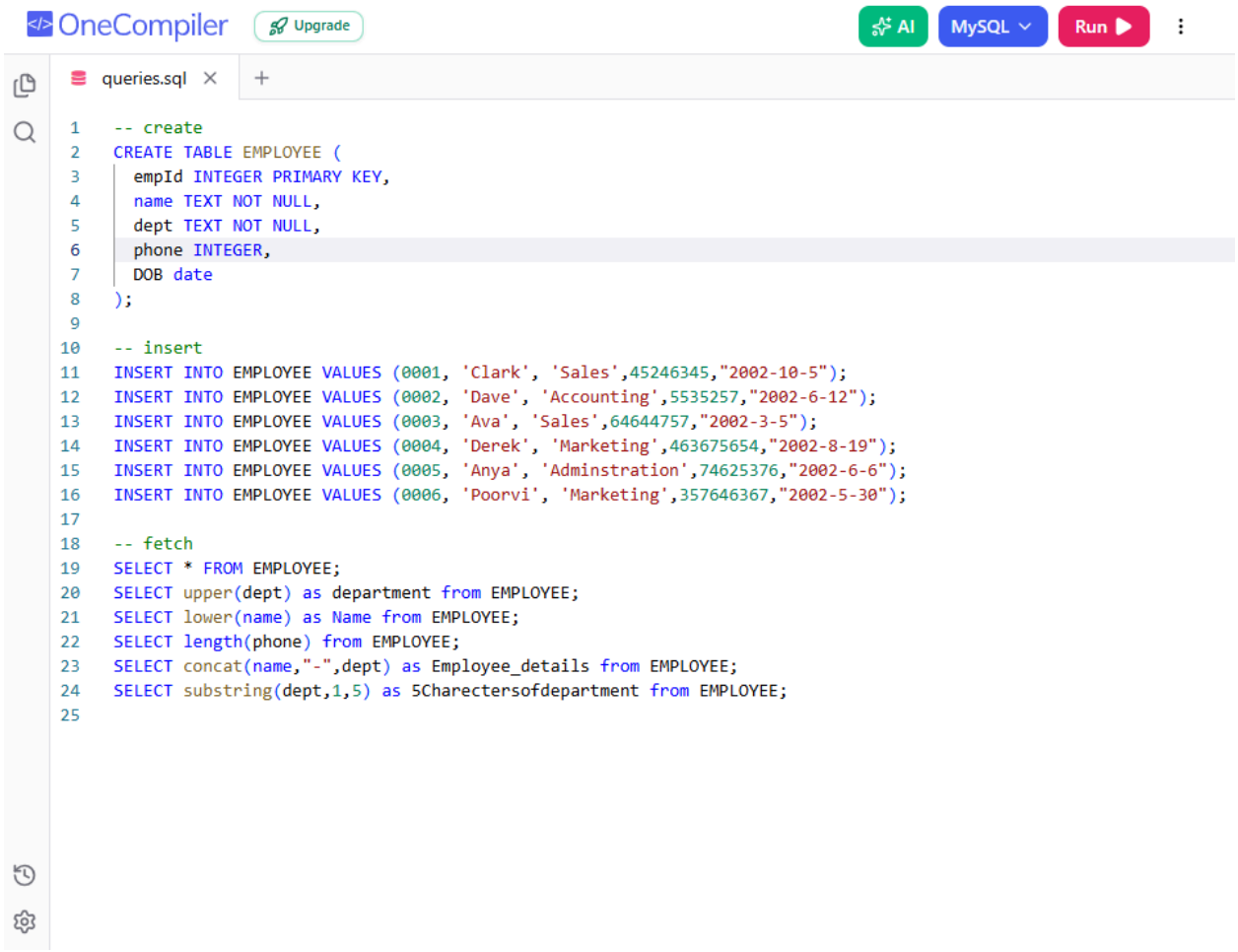



```sql
-- create
CREATE TABLE EMPLOYEE (
  empId INTEGER PRIMARY KEY,
  name TEXT NOT NULL,
  dept TEXT NOT NULL,
  phone INTEGER,
  DOB date
);

-- insert
INSERT INTO EMPLOYEE VALUES (0001, 'Clark', 'Sales',45246345,"2002-10-5");
INSERT INTO EMPLOYEE VALUES (0002, 'Dave', 'Accounting',5535257,"2002-6-12");
INSERT INTO EMPLOYEE VALUES (0003, 'Ava', 'Sales',64644757,"2002-3-5");
INSERT INTO EMPLOYEE VALUES (0004, 'Derek', 'Marketing',463675654,"2002-8-19");
INSERT INTO EMPLOYEE VALUES (0005, 'Anya', 'Adminstration',74625376,"2002-6-6");
INSERT INTO EMPLOYEE VALUES (0006, 'Poorvi', 'Marketing',357646367,"2002-5-30");

-- fetch
SELECT * FROM EMPLOYEE;
SELECT upper(dept) as department from EMPLOYEE;
SELECT lower(name) as Name from EMPLOYEE;
SELECT length(phone) from EMPLOYEE;
SELECT concat(name,"-",dept) as Employee_details from EMPLOYEE;
SELECT substring(dept,1,5) as 5Charectersofdepartment from EMPLOYEE;

```

Output:

empId	name	dept	phone	DOB
1	Clark	Sales	45246345	2002-10-05
2	Dave	Accounting	5535257	2002-06-12
3	Ava	Sales	64644757	2002-03-05
4	Derek	Marketing	463675654	2002-08-19
5	Anya	Adminstration	74625376	2002-06-06
6	Poorvi	Marketing	357646367	2002-05-30

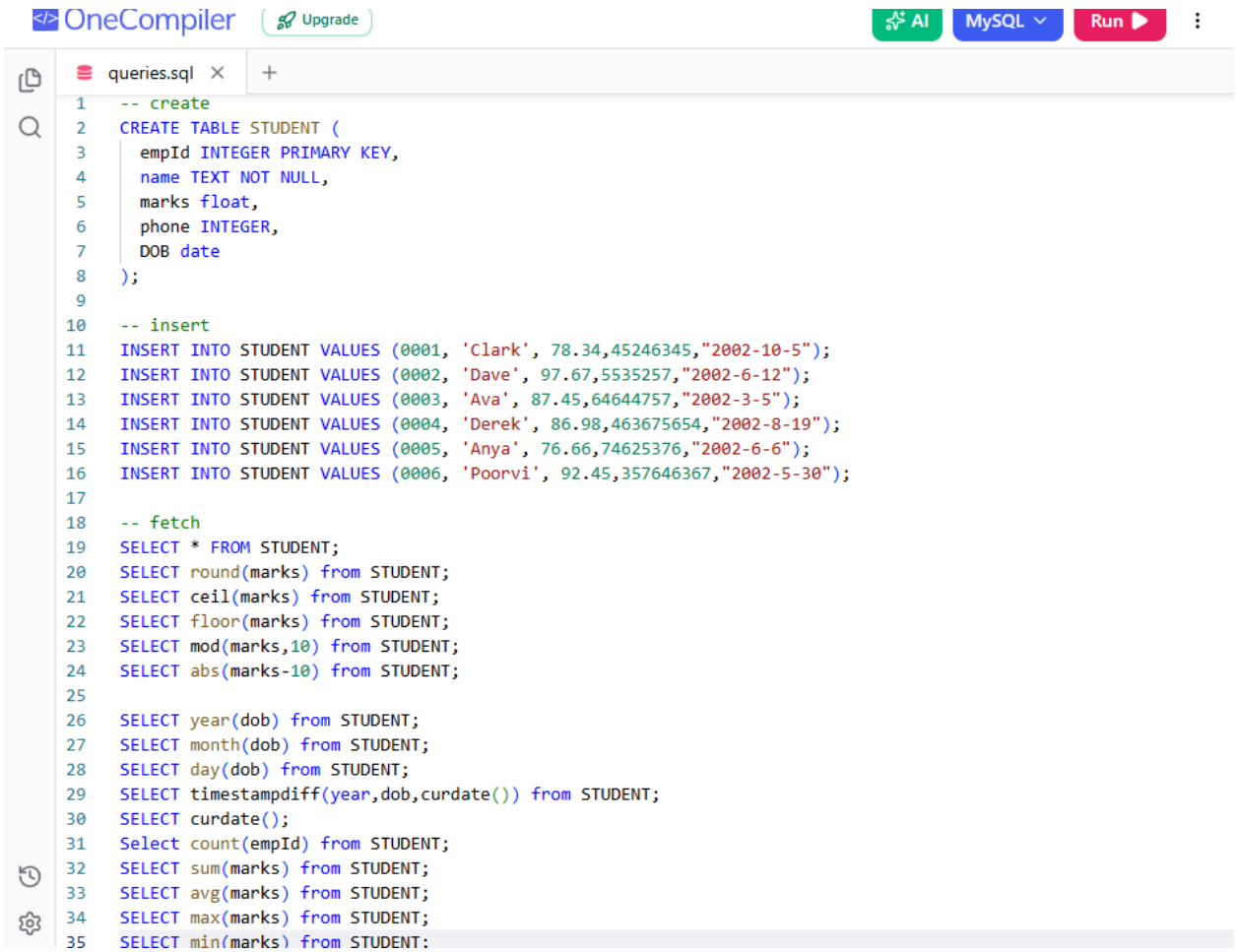
department
SALES
ACCOUNTING
SALES
MARKETING
ADMINSTRATION
MARKETING

```
+--------+
| Name   |
+--------+
| clark  |
| dave   |
| ava    |
| derek  |
| anya   |
| poorvi |
+--------+
```

```
+---------------+
| length(phone) |
+---------------+
|             8 |
|             7 |
|             8 |
|             9 |
|             8 |
|             9 |
+---------------+
```

```
+--------------------+
| Employee_details   |
+--------------------+
| Clark-Sales        |
| Dave-Accounting    |
| Ava-Sales          |
| Derek-Marketing    |
| Anya-Adminstration |
| Poorvi-Marketing   |
+--------------------+
```

```
+-------------------------+
| 5Charectersofdepartment |
+-------------------------+
| Sales                   |
| Accou                   |
| Sales                   |
| Marke                   |
| Admin                   |
| Marke                   |
+-------------------------+
```




```sql
-- create
CREATE TABLE STUDENT (
  empId INTEGER PRIMARY KEY,
  name TEXT NOT NULL,
  marks float,
  phone INTEGER,
  DOB date
);

-- insert
INSERT INTO STUDENT VALUES (0001, 'Clark', 78.34,45246345,"2002-10-5");
INSERT INTO STUDENT VALUES (0002, 'Dave', 97.67,5535257,"2002-6-12");
INSERT INTO STUDENT VALUES (0003, 'Ava', 87.45,64644757,"2002-3-5");
INSERT INTO STUDENT VALUES (0004, 'Derek', 86.98,463675654,"2002-8-19");
INSERT INTO STUDENT VALUES (0005, 'Anya', 76.66,74625376,"2002-6-6");
INSERT INTO STUDENT VALUES (0006, 'Poorvi', 92.45,357646367,"2002-5-30");

-- fetch
SELECT * FROM STUDENT;
SELECT round(marks) from STUDENT;
SELECT ceil(marks) from STUDENT;
SELECT floor(marks) from STUDENT;
SELECT mod(marks,10) from STUDENT;
SELECT abs(marks-10) from STUDENT;

SELECT year(dob) from STUDENT;
SELECT month(dob) from STUDENT;
SELECT day(dob) from STUDENT;
SELECT timestampdiff(year,dob,curdate()) from STUDENT;
SELECT curdate();
Select count(empId) from STUDENT;
SELECT sum(marks) from STUDENT;
SELECT avg(marks) from STUDENT;
SELECT max(marks) from STUDENT;
SELECT min(marks) from STUDENT;
```

Output:

empId	name	marks	phone	DOB
1	Clark	78.34	45246345	2002-10-05
2	Dave	97.67	5535257	2002-06-12
3	Ava	87.45	64644757	2002-03-05
4	Derek	86.98	463675654	2002-08-19
5	Anya	76.66	74625376	2002-06-06
6	Poorvi	92.45	357646367	2002-05-30

round(marks)
78
98
87
87
77
92

```
+--------------+
+-------------+
| ceil(marks) |
+-------------+
|          79 |
|          98 |
|          88 |
|          87 |
|          77 |
|          93 |
+-------------+
+--------------+
| floor(marks) |
+--------------+
|           78 |
|           97 |
|           87 |
|           86 |
|           76 |
|           92 |
+--------------+
+--------------------+
| mod(marks,10)      |
+--------------------+
|  8.339996337890625 |
```

```
+--------------------+
| mod(marks,10)      |
+--------------------+
|  8.339996337890625 |
| 7.6699981689453125 |
| 7.4499969482421875 |
|  6.980003356933594 |
|  6.660003662109375 |
| 2.4499969482421875 |
+--------------------+
+-------------------+
| abs(marks-10)     |
+-------------------+
| 68.33999633789062 |
| 87.66999816894531 |
| 77.44999694824219 |
|  76.9800033569336 |
| 66.66000366210938 |
| 82.44999694824219 |
+-------------------+
```

```
+-----------+
| year(dob) |
+-----------+
|      2002 |
|      2002 |
|      2002 |
|      2002 |
|      2002 |
|      2002 |
+-----------+
+------------+
| month(dob) |
+------------+
|         10 |
|          6 |
|          3 |
|          8 |
|          6 |
|          5 |
+------------+
```

Output:

```
| day(dob) |
+----------+
|        5 |
|       12 |
|        5 |
|       19 |
|        6 |
|       30 |
+----------+
```

```
+-----------------------------------+
| timestampdiff(year,dob,curdate()) |
+-----------------------------------+
|                                23 |
|                                24 |
|                                24 |
|                                24 |
|                                24 |
|                                24 |
+-----------------------------------+
```

```
+------------+
| curdate()  |
+------------+
| 2026-02-25 |
```

```
+------------+
| curdate()  |
+------------+
| 2026-08-25 |
+------------+
+--------------+
| count(empId) |
+--------------+
|            6 |
+--------------+
+-------------------+
| sum(marks)        |
+-------------------+
| 519.5499954223633 |
+-------------------+
+-------------------+
| avg(marks)        |
+-------------------+
| 86.59166590372722 |
+-------------------+
+------------+
| max(marks) |
+------------+
|      97.67 |
+------------+
```

```
+------------+
| max(marks) |
+------------+
|      97.67 |
+------------+
+------------+
| min(marks) |
+------------+
|      76.66 |
+------------+
```